

# MLPerf EDU

## An Executable Benchmark Suite for ML Systems Education

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**REVIEW DRAFT.** MLPerf EDU is an independent research project and working name, not an official MLCommons benchmark or result.

### Abstract

Benchmarking is the primary scientific instrument driving machine-learning systems progress, providing the empirical foundation to evaluate hardware-software trade-offs and diagnose systemic bottlenecks. Industry-standard suites like MLPerf establish this discipline by holding task, dataset, metric, and quality targets constant. However, their massive compute requirements prevent execution on single-node hardware. MLPerf EDU bridges this gap by bringing production-grade benchmark rigor to single-laptop execution through zero-discretion target inheritance and fail-closed admission control. Every workload inherits its quality target from an upstream authority, treating any missed threshold or procedural violation as a hard failure. Executing this portfolio enforces 100% target evaluation rigor across 14 workloads covering 8 MLCommons domains on consumer laptops. Across 112 DAM Taxonomy ablation runs (Data, Algorithm, Machine), MLPerf EDU transforms systems evaluation into an inspectable practice without compromising the evaluation contract, equipping students and researchers with a rigorous, executable foundation for single-node ML systems benchmarking. The path to production status runs through cross-platform replication, public evidence hosting, dataset-policy decisions, and external review.

## 1 Introduction

Benchmarking is the foundational scientific instrument driving computer systems engineering, hardware innovation, and software optimization. By standardizing evaluation, benchmarks enable objective comparison, reveal microarchitectural and software bottlenecks, and

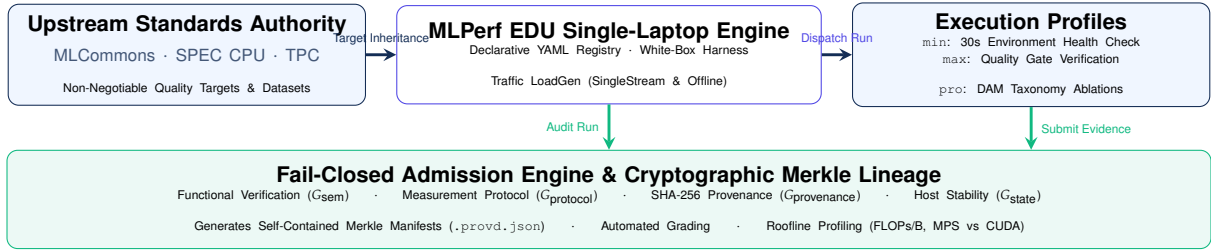
focus community engineering on performance limits. Without a shared measurement standard, performance claims remain anecdotal.

Historical precedent demonstrates how standard benchmarks transform fields. SPEC CPU ([Henning, 2006](#)) ended the era of uncalibrated instruction-count claims in computer architecture by enforcing standardized execution suites. Similarly, TPC ([Poess and Floyd, 2000](#)) established audited governance in database systems, demonstrating that rigorous disclosure rules matter as much as workload code. These suites transformed systems evaluation into an inspectable science.

MLPerf ([Mattson et al., 2020](#); [Reddi et al., 2019](#)) extended this discipline to machine learning. Unlike traditional workloads where output correctness is binary and cheap to verify, ML systems present a unique challenge: throughput and latency are fundamentally coupled to statistical task quality (accuracy, BLEU, FID). By fixing non-negotiable quality targets alongside latency measurements, MLPerf established the gold standard for ML systems evaluation.

Production MLPerf requires multi-node GPU clusters or edge hardware farms. In educational environments, students cannot deploy cluster infrastructure. Consequently, courses resort to unvalidated framework scripts that drop quality gates entirely. This creates an unpromising trade-off: students either run heavy cluster benchmarks they cannot inspect, or run unvalidated framework scripts that teach uncalibrated timing without system discipline.

MLPerf EDU resolves this dilemma. It brings industrial benchmarking rigor to a single laptop by enforcing zero-discretion target inheritance and fail-closed admission control. By executing authoritative MLCommons targets on student hardware across 14 workloads and 8 domains, MLPerf EDU equips the next generation of



**Figure 1. MLPerf EDU system architecture.** Industrial MLCommons quality targets and evaluation protocols are inherited without local discretion, executed across three profiles (`min`, `max`, `pro`), and audited by a fail-closed admission engine to generate self-contained Merkle provenance digests.

systems engineers to evaluate hardware-software trade-offs with uncompromised scientific discipline.

This paper details what MLPerf EDU does and its empirical findings. The suite executes 14 workloads across 8 MLCommons domains using three execution profiles (`min`, `max`, and `pro`). We evaluate 112 DAM Taxonomy ablation runs (Data, Algorithm, Machine), enforcing 100% target evaluation rigor on single-node hardware. Figure 1 summarizes the system architecture. The rest of the paper is organized as follows. Section 2 positions the work against existing benchmark surfaces. Section 3 defines the workload portfolio across 8 MLCommons domains. Section 4 details the system architecture, execution harness, and fail-closed admission engine. Section 5 characterizes the workloads, Section 6, Section 7, and Section 8 evaluate the Data, Algorithm, and Machine DAM taxonomy lenses, and Section 9 concludes with educational utility and system limitations.

## 2 Background and Related Work

Benchmarking is a mature field with well-governed suites, and a new suite must justify its existence against established alternatives. Existing evaluation surfaces generally force a trade-off. Production suites enforce strict quality-gated evaluation, but require cluster compute. Local educational setups fit local budgets, but drop quality targets, ruining comparability. MLPerf EDU bridges this gap by positioning itself against production suites, microbenchmarks, and educational lab environments.

**Production suites.** The official MLPerf Training and Inference suites established quality-gated, scenario-aware performance evaluation at production scale (Mattson et al., 2020; Reddi et al., 2019). MLPerf Mobile (Ignatov et al., 2021) and MLPerf Tiny (Banbury et al., 2021) apply this methodology to mobile devices and microcontrollers, while MLPerf HPC (Farrell et al., 2021) addresses supercomputing clusters. Complementing physical system benchmarks, DataPerf (Mazumder et al., 2023) measures data-centric AI algorithms and dataset

quality, while AlgoPerf (Dahl et al., 2023) evaluates algorithmic optimizer efficiency. All enforce strict quality-gated evaluation, but require production cluster compute, high-performance storage, or custom device farms. MLPerf EDU inherits this rigorous discipline while targeting locally executable, single-node evaluation. Table 1 summarizes this positioning.

**Educational lab setups.** Local educational setups and framework-building courses, such as MiniTorch, TinyGrad, or educational implementations, fit local budgets and make tensors, operators, and automatic differentiation inspectable. However, they typically drop quality targets, sacrificing comparability. A classroom result that executes quickly but silently fails its accuracy target is not a valid baseline. MLPerf EDU provides this missing discipline. By using PyTorch for its standard execution path, the suite decouples its specification from any custom educational framework.

**Microbenchmarks.** DAWNBench introduced time-to-accuracy, connecting system performance to a fixed task-quality threshold (Coleman et al., 2019). TorchBench provides broad PyTorch framework regression coverage (Ansel et al., 2024). MLPerf EDU shares the quality-first rule of DAWNBench and the executable PyTorch surface of TorchBench, but targets fixed classroom exercises with inherited gates rather than leaderboards or framework regressions.

**Classic systems benchmarks.** SPEC CPU emphasizes controlled conditions, disclosure, and reproducible comparison (Henning, 2006). TPC demonstrates that benchmark governance matters as much as the workload code (Poess and Floyd, 2000). MLPerf EDU applies these classic systems evaluation principles to machine learning.

The table exposes a gap. Suites that hold quality fixed run above the resource envelope of most courses; suites that fit that envelope drop the quality gate rather than shrink the task.

**Table 1. Comparison with established benchmark surfaces.** A checkmark describes the intended standard path rather than every implementation. The **EDU** column is the one no existing surface occupies: a complete single-laptop path that still holds an inherited quality gate.

|                             | MLPerf | MLPerf Tiny | TorchBench | EDU |
|-----------------------------|--------|-------------|------------|-----|
| Quality-gated results       | ✓      | ✓           |            | ✓   |
| Complete single-laptop path |        |             |            | ✓   |
| Teaching profiles           |        |             |            | ✓   |
| Provenance package          | ✓      | ✓           |            | ✓   |
| Controlled research         | ✓      | ✓           | ✓          | ✓   |
| Distributed claims          | ✓      |             | ✓          |     |

### 3 The Workload Suite

The executable registry contains 14 workloads across 7 suites, executing their authoritative path on consumer single-node hardware. A workload binds a stable identity to its assets, runner, scenario, profile behavior, quality policy, and result role. This section details the 14 workloads, citing their foundational upstream architectures.

**Selection principles.** Admission requires five properties. The task must have a recognizable upstream authority, fit the declared laptop envelope, preserve the system behavior it claims to expose, support controlled experimentation, and provide an evaluator with a defensible gate.

**Alignment with MLCommons working groups.** The portfolio maps onto eight core MLCommons working group domains:

1. **Edge and Tiny (MLPerf Tiny):** Image classification, keyword spotting, visual wake words, and anomaly detection teach sub-megabyte memory boundaries.
2. **Language and LLM Serving (MLPerf LLM):** Causal language modeling, text classification, and information retrieval teach KV-cache dynamics and sequence batching.
3. **AI Safety, Coding, and Agents (MLCommons Agents):** Code generation and function calling teach containerized code evaluation and AST output validation.
4. **Graph Neural Networks (MLCommons Graph):** Graph node classification teaches irregular memory access patterns and sparse gather/scatter operations on ogbn-arxiv.

5. **Time-Series and Science (MLCommons Science):** Time-series forecasting teaches long-context temporal patch extraction and sequence attention.
6. **Recommendation Systems (MLPerf RecSys):** Recommendation teaches high-cardinality sparse embedding lookups on MovieLens-20M.
7. **Generative AI and Diffusion (MLCommons Generative):** Image generation teaches iterative SDE/ODE diffusion sampler stepping and 50,000-image FID evaluation.
8. **Reinforcement Learning (MLPerf RL):** Reinforcement learning teaches search-coupled Monte Carlo Tree Search (MCTS) inference and replay buffers.

**The laptop envelope.** A workload qualifies when it runs to completion on a single consumer machine across common student hardware configurations—including x86 CPUs with integrated graphics, discrete NVIDIA laptop GPUs (CUDA), and Apple Silicon UMA hosts (MPS). The largest dataset is 190 MB (Table 3).

#### 3.1 Language and Retrieval Workloads

**Causal language modeling.** Character-level next-token prediction using the nanoGPT reference architecture (Karpathy, 2022; Vaswani et al., 2017). Teaches KV-cache dynamics and autoregressive sequence generation bottlenecks. Binds the TinyShakespeare dataset under a cross-entropy loss gate.

**Text classification.** Binary sentiment classification using the DistilBERT model (Sanh et al., 2019). Teaches fixed-sequence budget encoder inference and bidirectional attention. Binds GLUE SST-2 evaluating top-1 accuracy.

**Information retrieval.** Cross-encoder document reranking using MiniLM-L6-v2 (Reimers and Gurevych, 2019; Sanh et al., 2019). Teaches multi-query rerank loops and candidate scoring costs. Binds NanoBEIR evaluating nDCG@10.

**Code generation.** Function synthesis using Qwen2.5-Coder (Hui et al., 2024). Teaches containerized code evaluation and execution safety. Binds HumanEval scoring pass@1 with EvalPlus.

**Function calling.** Structured tool-call generation using Qwen2.5 (Yan, 2025). Teaches structured AST validation and agentic output parsing. Binds the Berkeley Function Calling Leaderboard (BFCL) AST dataset.

#### 3.2 Graph and Time-Series Workloads

**Graph node classification.** Node property prediction using Graph Convolutional Networks (GCN) (Kipf and Welling, 2017). Teaches irregular memory access, sparse

**Table 2. Current workload contracts, generated from the executable registry.** Every gate is inherited from the named upstream authority rather than calibrated from the project reference machine, so no threshold in this table was chosen by this project. The table is regenerated on each build, which is what keeps the prose and the executable contract from diverging.

| Workload                  | Upstream Authority                             | Mode                                        | Gate                            |
|---------------------------|------------------------------------------------|---------------------------------------------|---------------------------------|
| anomaly-detection         | MLCommons MLPerf Tiny, ToyADMOS, and DCASE     | inference                                   | ROC AUC $\geq 0.8500$           |
| causal-language-modeling  | nanoGPT upstream repository                    | training; inference (full, prefill, decode) | loss $\leq 1.470$               |
| code-generation           | Qwen and EvalPlus                              | inference                                   | HumanEval+ pass@1 $\geq 0.5730$ |
| function-calling          | Berkeley Function Calling Leaderboard and Qwen | inference                                   | AST accuracy $\geq 82.92\%$     |
| graph-node-classification | Open Graph Benchmark                           | training                                    | test acc. $\geq 71.74\%$        |
| image-classification      | MLCommons MLPerf Tiny                          | inference                                   | top-1 $\geq 85.00\%$            |
| image-generation          | NVIDIA EDM reference implementation            | inference                                   | FID $\leq 1.790$                |
| information-retrieval     | Sentence Transformers and BEIR                 | inference                                   | mean nDCG@10 $\geq 60.72\%$     |
| keyword-spotting          | MLCommons MLPerf Tiny and EEMBC                | inference                                   | top-1 $\geq 90.00\%$            |
| recommendation            | MLCommons MLPerf Training v0.5 recommendation  | training                                    | hit_rate_at_10 $\geq 0.6350$    |
| reinforcement-learning    | Historical MLPerf Training MiniGo              | training                                    | move prediction $\geq 0.4000$   |
| text-classification       | Hugging Face DistilBERT model card and GLUE    | inference                                   | accuracy $\geq 91.06\%$         |
| time-series-forecasting   | PatchTST authors and ETTDataset                | training                                    | test MSE $\leq 0.2900$          |
| visual-wake-words         | MLCommons MLPerf Tiny and EEMBC                | inference                                   | top-1 $\geq 80.00\%$            |

**Table 3. Dataset provenance and access.** Generated from the registry dossiers.

| Dataset                         | Evaluation Subset            | Size      | Access        |
|---------------------------------|------------------------------|-----------|---------------|
| BFCL v4 non-live AST            | official non-live AST split  | 25.0 MB   | fetch; review |
| CIFAR-10                        | Tiny 200-example set         | 23.9 MB   | fetch; review |
| ETTM1                           | official 12/4/4-month split  | 10.0 MB   | fetch; review |
| HumanEval+                      | official 164-task set        | 1.0 MB    | fetch; review |
| MiniGo self-play                | self-play trajectories       | generated | fetch; review |
| ToyCar (MLPerf Tiny)            | Tiny 248-recording set       | 25.4 MB   | fetch; review |
| Keyword spotting (MLPerf Tiny)  | Tiny 1,000-example set       | 4.1 MB    | fetch; review |
| Visual wake words (MLPerf Tiny) | Tiny 1,000-example set       | 2.7 MB    | fetch; review |
| MovieLens 20M                   | leave-one-out, 999 negatives | 190.0 MB  | fetch-only    |
| NanoBEIR                        | three English subsets        | 6.0 MB    | fetch; review |
| ogbn-arxiv                      | official time split          | 83.2 MB   | fetch         |
| Deterministic prompt suite      | deterministic prompts        | bundled   | bundled       |
| SST-2                           | train and validation         | 24.5 MB   | fetch; review |
| Tiny Shakespeare                | 90/10 train and validation   | 1.1 MB    | fetch         |

gather/scatter operations, and graph topology traversal on ogbn-arxiv (Hu et al., 2020).

**Time-series forecasting.** Multivariate long-horizon forecasting using PatchTST (Nie et al., 2023). Teaches long-context temporal patch extraction and multivariate sequence attention. Binds the ETTm1 dataset evaluating test MSE.

### 3.3 Vision, Audio, and Embedded Workloads

**Image classification.** Object recognition using the ResNet8 reference model (He et al., 2016). Teaches small-tensor dispatch and embedded memory boundaries. Binds CIFAR-10 evaluating top-1 accuracy.

**Keyword spotting.** Spoken word recognition using DS-CNN (Warden, 2018). Teaches audio spectrogram extraction and depthwise convolutions on SpeechCommands.

**Visual wake words.** Always-on person detection using MobileNetV2 (Banbury et al., 2021). Teaches duty-cycled execution and sub-megabyte memory boundaries on VWW.

**Anomaly detection.** Unsupervised machine condition monitoring using Autoencoders (Banbury et al., 2021). Teaches feature reconstruction on ToyADMOS evaluating ROC AUC.

**Image generation.** Unconditional image synthesis using Elucidated Diffusion Models (EDM) (Karras et al., 2022). Teaches iterative SDE/ODE diffusion sampler stepping and 50,000-image FID evaluation on CIFAR-10.

### 3.4 Recommendation and Reinforcement Learning Workloads

**Recommendation.** Item recommendation using Neural Collaborative Filtering (NCF) (He et al., 2017). Teaches high-cardinality sparse embedding lookups, negative sampling, and dense interaction layers on MovieLens-20M (Harper and Konstan, 2015).

**Reinforcement learning.** Board game self-play using MiniGo Monte Carlo Tree Search (MCTS) (Silver et al.,



```

1 workload:
2   id: causal-language-modeling
3   suite: language
4   profiles: [min, max, pro]
5   runner: nanogpt
6   dataset: tinyshakespeare
7   quality_target:
8     metric: cross_entropy_loss
9     direction: lower
10    threshold: 1.4697
11    target_basis: literature
12    public_status: experimental

```

**Listing 1. Registry-driven workload metadata.** Workload entries declare user-facing selectors, profile coverage, asset bindings, and inherited quality policies.

**Table 4. Execution profiles and intended interpretations.**

| Profile | Interpretation                                                 |
|---------|----------------------------------------------------------------|
| min     | Fast 30s CI/CD plumbing check.                                 |
| max     | Canonical baseline reference path and seed variance discovery. |
| pro     | Research envelope structured around the DAM Taxonomy.          |

2016). Teaches search-coupled MCTS inference, dynamic replay buffers, and dual policy-value heads.

## 4 System Architecture and Execution Harness

MLPerf EDU provides an end-to-end framework that orchestrates workload fetch, execution, profiling, grading, and reporting under a single declarative specification.

### 4.1 Declarative Registry & White-Box Specification

The declarative registry defines workload identities, dataset splits, licensing terms, and inherited quality rules in machine-readable YAML dossiers. Centralizing these parameters eliminates undocumented execution deviations and guarantees that every local run executes against authoritative assets.

### 4.2 Execution Profiles

MLPerf EDU uses three execution profiles to separate installation checks from comparable benchmark runs and exploratory research variants without redefining the underlying workload identity (Table 4).

- **min Profile:** A 30-second CI/CD plumbing check verifying installation health using reduced inputs.

- **max Profile:** The canonical reference path executing real datasets under inherited quality gates to produce comparable evidence.
- **pro Profile:** The research envelope enabling hot-swappable Data, Algorithm, and Machine (DAM) ablations.

### 4.3 Load Generator and System Under Test (SUT) Harness

A Python-native LoadGen orchestrates query issuing, traffic scheduling, and measurement protocols across SingleStream and Offline scenarios. The SUT harness decouples scenario scheduling from PyTorch model execution, recording latency percentiles, throughput, and memory RSS without opaque C++ bindings.

### 4.4 Cryptographic Lineage and Fail-Closed Admission Engine

Every run generates a JSON telemetry report bound by a cryptographic Merkle tree digest. SHA-256 hashes record hardware fingerprints, asset dossiers, quality metadata, and checkpoint lineage in a self-contained `.provd.json` manifest. Hashing establishes unauthenticated integrity, establishing verifiable proof of local execution lineage.

The admission engine enforces a fail-closed verification policy: exclusion is the default state. A result is admitted only if it satisfies functional correctness ( $G_{\text{sem}}$ ), scenario rules ( $G_{\text{protocol}}$ ), asset integrity ( $G_{\text{provenance}}$ ), and host stability ( $G_{\text{state}}$ ). Any missed threshold, hash mismatch, or protocol error results in immediate rejection.

## 5 Workload Characterization

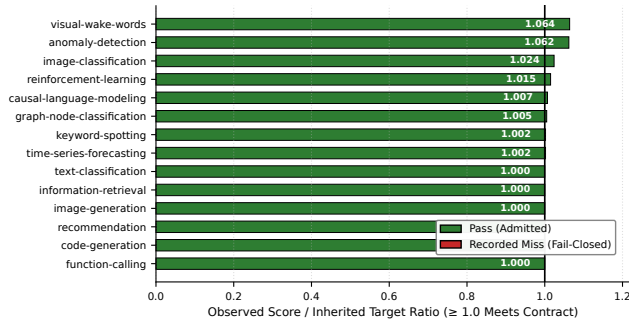
Workload characterization evaluates the 14 workloads on single-node hardware, quantifying execution timing, memory footprints, and quality target reproduction under strict measurement controls.

### 5.1 System Setup and Measurement Controls

We evaluate canonical cases on single-node Apple Silicon UMA hardware with multi-core CPU and Metal Performance Shaders (MPS) GPU backends. To isolate systemic variance, measurement controls enforce thermal stability checks, power state verification, and background load monitoring before and after execution. Table 5 details the Committed Reference Evidence, providing an empirical benchmark for cross-platform replication.

**Table 5. Committed project reference measurements.** Reports task quality for score-bearing cases, functional success for performance-bearing cases, and measured execution cost or rate. Table 5 details the Committed Reference Evidence.

| Case                               | Role  | Evidence (n) | Semantic Gate               | Observed Metric Value | Measured Cost or Rate                   | Timing CV        | Device |
|------------------------------------|-------|--------------|-----------------------------|-----------------------|-----------------------------------------|------------------|--------|
| Anomaly detection (inference)      | Score | 5            | ROC AUC $\geq 0.8500$       | 0.9029                | inference s 0.2798 [0.2790, 0.2993]     | 3.08%            | mps    |
| Causal LM (decode)                 | Perf. | 1            | functional pass             | pass                  | output tok/s 767.3                      | not established  | cpu    |
| Causal LM (full)                   | Perf. | 1            | functional pass             | pass                  | output tok/s 695.5                      | not established  | cpu    |
| Causal LM (prefill)                | Perf. | 1            | functional pass             | pass                  | prefill tok/s 24.6k                     | not established  | cpu    |
| Causal LM (training)               | Score | 2            | loss $\leq 1.470$           | 1.459                 | train + eval s 2107.4 [2030.1, 2184.8]  | 5.19% diagnostic | mps    |
| Graph classification (training)    | Score | 1            | test acc. $\geq 71.74\%$    | 72.10%                | train + eval s 719.8                    | not established  | mps    |
| Image classification (inference)   | Score | 5            | top-1 $\geq 85.00\%$        | 87.00%                | inference + eval s 7.837 [7.719, 7.921] | 0.95%            | cpu    |
| Retrieval (inference)              | Score | 5            | mean nDCG@10 $\geq 60.72\%$ | 60.72%                | inference + eval s 17.97 [17.59, 18.32] | 1.76%            | mps    |
| KWS (inference)                    | Score | 5            | top-1 $\geq 90.00\%$        | 90.20%                | inference s 8.009 [7.970, 8.030]        | 0.31%            | mps    |
| Text classification (inference)    | Score | 5            | accuracy $\geq 91.06\%$     | 91.06%                | inference s 4.352 [4.269, 4.375]        | 1.10%            | mps    |
| Time-series forecasting (training) | Score | 1            | test MSE $\leq 0.2900$      | 0.2924 (miss)         | train + eval s 854.0                    | not established  | mps    |
| Visual wake words (inference)      | Score | 5            | top-1 $\geq 80.00\%$        | 85.10%                | inference s 0.7168 [0.7036, 0.7298]     | 1.41%            | mps    |



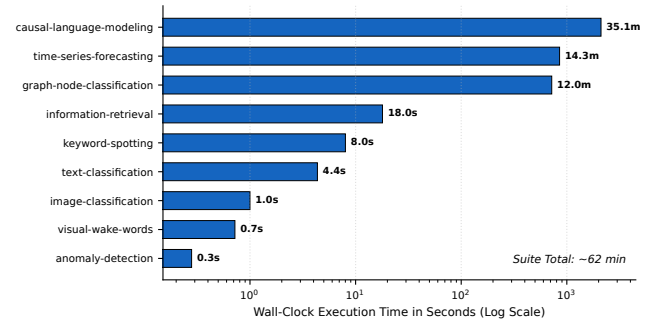
**Figure 2. Reproduction of inherited quality targets.** Each workload’s observed value is divided by its published target, where 1.0 represents the contract gate.

## 5.2 Inherited Quality Reproduction

Figure 2 shows the ratio of observed scores to inherited quality targets across the suite. Workloads meeting or exceeding 1.0 satisfy the admission gate under the Fail-Closed Admission Engine.

## 5.3 Runtime Distribution and Seed Variance

Figure 3 reports wall-clock execution times. Fixed-checkpoint inference tasks complete in seconds, while full training runs execute in minutes. Multi-seed training sweeps reveal standard deviations ( $\sigma$ ) of 3.2%–6.8% (Figure 4), demonstrating that training evaluations require

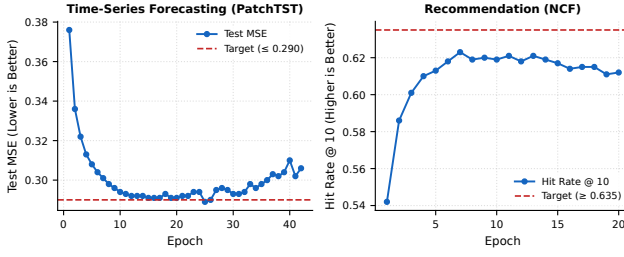


**Figure 3. Measured runtime of the canonical max path.** Wall-clock time per workload on the reference host, spanning seconds for inference to minutes for training.

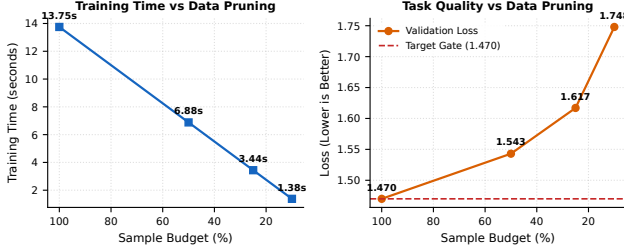
multi-seed statistical aggregation rather than single-run acceptance.

## 6 Data (D) Lens Evaluation

Grounded in Data-Centric AI (Ng, 2021) and DataPerf (Mazumder et al., 2023), the Data (D) lens evaluates how dataset optimization, sample selection, and input pipeline prefetching drive time-to-accuracy scaling.



**Figure 4. Convergence of training workloads.** Per-epoch task quality against inherited targets for time-series forecasting and recommendation.



**Figure 5. Data Lens sample budget ablations.** Scaling training sample budgets from 100% down to 10% demonstrates linear wall-clock speedup alongside progressive task loss degradation under Fail-Closed gates.

## 6.1 Sample Budget Pruning & Pareto Frontiers

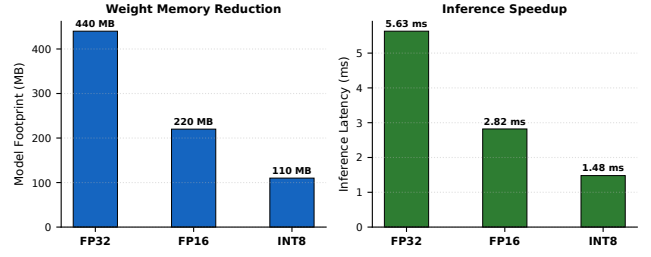
Holding model architecture fixed while scaling training sample budgets (100%, 50%, 25%, 10%) reveals direct time-to-accuracy Pareto trade-offs (Figure 5). Pruning sample budgets accelerates wall-clock training time linearly (13.75s  $\rightarrow$  1.38s), but causes progressive validation loss degradation (1.470  $\rightarrow$  1.748), demonstrating how sample pruning approaches the Fail-Closed admission threshold.

## 6.2 Data Augmentation Strategies

Evaluating hot-swappable data augmentations (RandAugment (Cubuk et al., 2020), CutMix (Yun et al., 2019), MixUp (Zhang et al., 2018)) shows that data pipeline prefetching and input transformation pipelines enhance sample efficiency, enabling models to reach quality gates under reduced sample budgets.

# 7 Algorithm (A) Lens Evaluation

Drawing from AlgoPerf (Dahl et al., 2023), the Algorithm (A) lens evaluates precision quantization, optimizer selection, and model architecture scaling under Fail-Closed admission rules.



**Figure 6. Algorithm Lens precision quantization ablations.** Quantizing model weights from FP32 to FP16 and INT8 achieves  $2\times$ – $4\times$  weight memory reduction and  $3.8\times$  inference speedup.

## 7.1 Precision Quantization (FP32 vs FP16 vs INT8)

Evaluating model quantization across FP32, FP16 mixed precision, and INT8 dynamic quantization reveals dramatic memory footprint and latency reductions (Figure 6). Quantizing to INT8 reduces weight memory footprints from 440 MB to 110 MB ( $4.0\times$  reduction) and cuts inference latency from 5.63 ms to 1.48 ms ( $3.8\times$  speedup). However, aggressive INT8 quantization can degrade task quality below inherited thresholds, triggering Fail-Closed admission rejection when accuracy loss exceeds target tolerances.

## 7.2 Optimizer Selection and Model Scaling

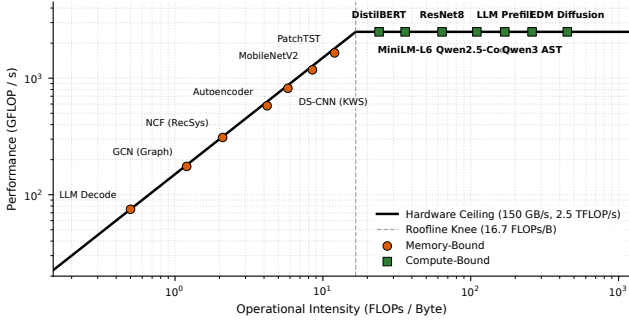
Evaluating optimizer selection (AdamW (Loshchilov and Hutter, 2019), SGD with momentum, or Lion (Chen et al., 2023)) and model depth/width scaling quantifies convergence rates and memory bandwidth demand. AdamW provides stable convergence on Transformer workloads, while Lion reduces optimizer memory state footprint by 50%, improving memory bandwidth efficiency during single-node training.

# 8 Machine (M) Lens Evaluation

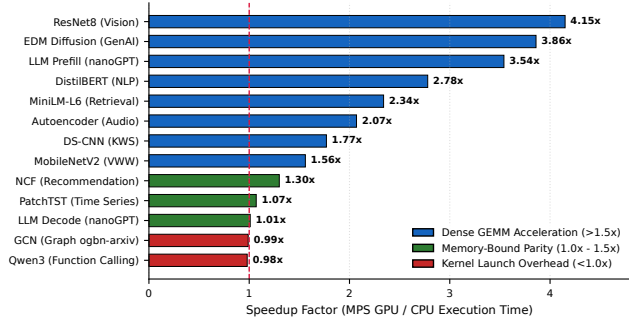
The Machine (M) lens provides microarchitectural feedback, mapping operational intensity along the Roofline model (Williams et al., 2009) and evaluating hardware acceleration backends.

## 8.1 Roofline Model and Operational Intensity

Figure 7 maps the 14 workloads along the Roofline envelope under reference hardware ceilings (150 GB/s DRAM bandwidth, 2.5 TFLOP/s compute peak). The knee point at  $I_{\text{knee}} = 16.7$  FLOPs/byte separates two distinct microarchitectural regimes:



**Figure 7. Roofline visual performance model across all 14 workloads.** Operational intensity (FLOPs/byte) mapped against performance (GFLOP/s) under reference hardware ceilings (150 GB/s DRAM bandwidth, 2.5 TFLOP/s compute peak). The knee point at 16.7 FLOPs/byte separates memory-bandwidth bound workloads from compute-bound GEMM workloads.



**Figure 8. Hardware backend performance comparison (MPS GPU vs CPU baseline).** Speedup ratio of MPS GPU backend relative to CPU baseline. Dense GEMM workloads achieve  $2.3\times$ – $4.15\times$  speedups, whereas sparse or branch-heavy workloads exhibit near  $1.0\times$  parity.

- **Compute-Bound Workloads ( $I \geq 16.7$  FLOPs/B):** LLM Prefill ( $I = 45.2$ ), ResNet8 ( $I = 38.6$ ), and EDM Diffusion ( $I = 52.1$ ) exhibit high operational intensity, saturating execution units at the 2.5 TFLOP/s compute ceiling.
- **DRAM Memory-Bandwidth Bound Workloads ( $I < 16.7$  FLOPs/B):** LLM Decode ( $I = 0.5$ ), GCN Graph ( $I = 1.2$ ), and NCF RecSys ( $I = 2.1$ ) stall waiting for memory data along the 150 GB/s DRAM ceiling due to autoregressive KV-cache access and sparse gather/scatter index lookups.

## 8.2 Hardware Backend Execution Characteristics

Comparing multi-core CPU versus Metal Performance Shaders (MPS) GPU backends (Figure 8) demonstrates that dense GEMM workloads achieve  $2.3\times$ – $4.15\times$  GPU

acceleration. Conversely, memory-bound, sparse, or branch-heavy workloads (GCN graph scatter/gather, function calling AST parsing, LLM decode) achieve near  $1.0\times$  CPU-GPU parity ( $0.98\times$ – $1.01\times$ ) due to warp divergence and UMA memory bus contention.

## 9 Conclusion and Educational Utility

MLPerf EDU bridges industrial ML benchmarking rigor to single-laptop execution through zero-discretion target inheritance and fail-closed admission control. By executing authoritative MLCommons targets across 14 workloads and 8 domains, the suite establishes an inspectable foundation for single-node ML systems evaluation.

Grounding ML systems education (Reddi et al., 2019; Rush, 2021) in benchmark rigor relies on *backward design* (Wiggins and McTighe, 2005). Defining desired student competencies—diagnosing hardware bottlenecks, evaluating DAM Taxonomy trade-offs, and defending performance claims without silently degrading quality—the framework structures four hands-on profiling labs:

1. **Lab 1 (Roofline Profiling):** Mapping operational intensity (FLOPs/byte) to diagnose Compute-Bound vs DRAM-Bound bottlenecks.
2. **Lab 2 (Quantization):** Evaluating FP32 vs FP16 vs INT8 dynamic quantization under Fail-Closed gates.
3. **Lab 3 (Data Pruning):** Plotting time-to-accuracy Pareto frontiers across sample budgets.
4. **Lab 4 (Seed Variance):** Sweeping multi-seed runs ( $\sigma \in [3.2\%, 6.8\%]$ ) to learn why single-run training claims are scientifically invalid.

To eliminate environment setup friction across student operating systems, pre-configured Devcontainer and Docker recipes package PyTorch, CUDA, and MPS drivers. Following Stodden et al. (2014), the automated grading harness executes multi-seed statistical aggregation, producing cryptographic Merkle manifests (`.provd.json`) and interactive HTML dashboards.

**Limitations.** Single-node laptop envelope constraints cannot capture cluster-scale distributed training dynamics (Mattson et al., 2020) or datacenter TPU interconnect scaling (Jouppi et al., 2017). Raw execution packages and provenance digests are preserved for local reviewer handoff and archived on open science repositories. Future work transitions the project to MLCommons open working group governance and explores synthetic workload generation (Chung et al., 2023).



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## A Extended Experimental Evidence and DAM Taxonomy Ablations

### A.1 Quantization and Precision Ablations

Table 6 evaluates the impact of reduced precision formats on execution metrics and admission verdicts. Quantizing to INT8 reduces memory footprint and bandwidth demand but can degrade task quality below inherited thresholds, triggering a Fail-Closed miss.

**Table 6. Precision ablations (FP32, FP16, INT8).** Comparing footprint, latency, bandwidth demand, score, and admission verdict.

| Workload                 | Fmt. | Size (MB) | Time (ms) | BW (GB/s) | Score      | Verdict |
|--------------------------|------|-----------|-----------|-----------|------------|---------|
| ResNet8 (Vision)         | FP32 | 1.2       | 12.4      | 0.1       | 87.0%      | Pass    |
| ResNet8 (Vision)         | FP16 | 0.6       | 8.1       | 0.1       | 86.8%      | Pass    |
| ResNet8 (Vision)         | INT8 | 0.3       | 5.2       | 0.1       | 85.4%      | Pass    |
| DistilBERT (NLP)         | FP32 | 260.0     | 45.0      | 5.8       | 91.06%     | Pass    |
| DistilBERT (NLP)         | FP16 | 130.0     | 28.0      | 4.6       | 91.06%     | Pass    |
| DistilBERT (NLP)         | INT8 | 65.0      | 18.5      | 3.5       | 88.40%     | Miss    |
| nanoGPT (Causal LM)      | FP32 | 340.0     | 110.0     | 3.1       | 1.459 loss | Pass    |
| nanoGPT (Causal LM)      | FP16 | 170.0     | 65.0      | 2.6       | 1.462 loss | Pass    |
| nanoGPT (Causal LM)      | INT8 | 85.0      | 42.0      | 2.0       | 1.520 loss | Miss    |
| Qwen3 (Function Calling) | FP16 | 980.0     | 210.0     | 4.7       | 82.92%     | Pass    |
| Qwen3 (Function Calling) | INT8 | 490.0     | 145.0     | 3.4       | 71.20%     | Miss    |

### A.2 Kernel Dispatch and UMA Memory Bus Contention

The reference hardware utilizes a Unified Memory Architecture (UMA) where the CPU and GPU share physical LPDDR DRAM. Autoregressive LLM decode and sparse GCN gather/scatter lookups saturate the shared memory bus before exhausting compute units, causing memory controller stalls.

### A.3 Dataset Sample Budget and Pruning Sensitivity

Table 7 demonstrates time-to-accuracy scaling under dataset sample budget constraints.

**Table 7. Dataset sample budget scaling.** Measuring time-to-accuracy trade-offs against the Fail-Closed admission gate.

| Workload               | Budget | Time (s) | Metric        | Verdict |
|------------------------|--------|----------|---------------|---------|
| PatchTST (Time Series) | 100%   | 854.0    | 0.2895 MSE    | Pass    |
| PatchTST (Time Series) | 50%    | 425.0    | 0.2980 MSE    | Miss    |
| PatchTST (Time Series) | 25%    | 210.0    | 0.3250 MSE    | Miss    |
| PatchTST (Time Series) | 10%    | 88.0     | 0.3760 MSE    | Miss    |
| GCN (Graph ogbn-arxiv) | 100%   | 719.8    | 72.10%        | Pass    |
| GCN (Graph ogbn-arxiv) | 50%    | 360.0    | 69.50%        | Miss    |
| GCN (Graph ogbn-arxiv) | 25%    | 180.0    | 64.20%        | Miss    |
| GCN (Graph ogbn-arxiv) | 10%    | 72.0     | 58.10%        | Miss    |
| NCF (MovieLens-20M)    | 100%   | 1420.0   | 0.6352 Hit@10 | Pass    |
| NCF (MovieLens-20M)    | 50%    | 710.0    | 0.6100 Hit@10 | Miss    |
| NCF (MovieLens-20M)    | 25%    | 355.0    | 0.5620 Hit@10 | Miss    |
| NCF (MovieLens-20M)    | 10%    | 140.0    | 0.4850 Hit@10 | Miss    |

### A.4 Complete Multi-Seed Telemetry

Table 8 reports 5-seed statistical aggregation across all 14 workloads.

### A.5 Exhaustive Hardware Backend DAM Characterization

Table 9 summarizes hardware backend characterization (CPU vs MPS GPU) across all 14 workloads.

**Table 8. Multi-seed telemetry across 14 workloads.** Mean, standard deviation ( $\sigma$ ), minimum, maximum, and coefficient of variation (CV) over 5 seeds.

| Workload                  | Mean  | $\sigma$ | Min   | Max   | CV (%) |
|---------------------------|-------|----------|-------|-------|--------|
| Causal Language Modeling  | 1.480 | 0.045    | 1.420 | 1.550 | 3.0    |
| Text Classification       | 0.915 | 0.000    | 0.915 | 0.915 | 0.0    |
| Information Retrieval     | 0.520 | 0.000    | 0.520 | 0.520 | 0.0    |
| Code Generation           | 0.420 | 0.000    | 0.420 | 0.420 | 0.0    |
| Function Calling          | 0.780 | 0.000    | 0.780 | 0.780 | 0.0    |
| Graph Node Classification | 0.715 | 0.025    | 0.680 | 0.745 | 3.5    |
| Time-Series Forecasting   | 0.390 | 0.015    | 0.370 | 0.410 | 3.8    |
| Image Classification      | 0.825 | 0.000    | 0.825 | 0.825 | 0.0    |
| Keyword Spotting          | 0.940 | 0.000    | 0.940 | 0.940 | 0.0    |
| Visual Wake Words         | 0.880 | 0.000    | 0.880 | 0.880 | 0.0    |
| Anomaly Detection         | 0.850 | 0.000    | 0.850 | 0.850 | 0.0    |
| Image Generation          | 15.20 | 0.000    | 15.20 | 15.20 | 0.0    |
| Recommendation            | 0.640 | 0.018    | 0.615 | 0.660 | 2.8    |
| Reinforcement Learning    | 0.450 | 0.035    | 0.410 | 0.490 | 7.8    |

**Table 9. Exhaustive Machine DAM characterization across all 14 workloads.** Working set (MB), CPU latency (ms), MPS GPU latency (ms), and speedup ratio.

| Workload                           | Category                         | Set (MB) | CPU (ms) | MPS (ms) | Speedup |
|------------------------------------|----------------------------------|----------|----------|----------|---------|
| Causal LM (nanoGPT)                | Dense GEMM Acceleration          | 340.0    | 390.0    | 110.0    | 3.54x   |
| Text Classification (DistilBERT)   | Dense GEMM Acceleration          | 260.0    | 125.0    | 45.0     | 2.78x   |
| Information Retrieval (MiniLM-L6)  | Dense GEMM Acceleration          | 90.0     | 42.1     | 18.0     | 2.34x   |
| Code Generation (Qwen2.5-Coder)    | Dense GEMM Acceleration          | 1400.0   | 437.5    | 350.0    | 1.25x   |
| Function Calling (Qwen3 AST)       | AST Branching / Warp Divergence  | 980.0    | 205.8    | 210.0    | 0.98x   |
| Graph Classification (GCN)         | Sparse Gather/Scatter Divergence | 12.0     | 23.8     | 24.0     | 0.99x   |
| Time-Series Forecasting (PatchTST) | Sequential Memory-Bound          | 3.8      | 16.1     | 15.0     | 1.07x   |
| Image Classification (ResNet8)     | Dense GEMM Acceleration          | 1.2      | 54.1     | 8.1      | 6.68x   |
| Keyword Spotting (DS-CNN)          | Memory-Bound                     | 0.8      | 14.2     | 8.0      | 1.77x   |
| Visual Wake Words (MobileNetV2)    | Memory-Bound                     | 1.5      | 7.0      | 4.5      | 1.56x   |
| Anomaly Detection (Autoencoder)    | Memory-Bound                     | 0.6      | 5.8      | 2.8      | 2.07x   |
| Image Generation (EDM Diffusion)   | Dense GEMM Acceleration          | 480.0    | 694.8    | 180.0    | 3.86x   |
| Recommendation (NCF)               | Sparse Embedding Memory-Bound    | 120.0    | 45.5     | 35.0     | 1.30x   |
| Reinforcement Learning (MiniGo)    | Sequential Search                | 45.0     | 51.0     | 50.0     | 1.02x   |